Assignment 9: Advanced SQL Queries Using Joins and Subqueries

ProjectRapido – Bike & Cab Booking System
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ProgrammeBCA Semester 3 (2025)
CourseDatabase Management Systems
File nameAssignment_9_Advanced_SQL_Queries.pdf

1. Objective

Produce management reports for Rapido that need more than one table: inner joins across riders, captains and vehicles; a left join to spot unpaid rides; subqueries for above-average captains and high-spending riders; and zone coverage plus vehicle-type revenue.

2. Report A — Complete booking sheet (four-table inner join)

Operations needs one row per ride with rider name, captain name and vehicle type, not raw IDs.

```
SELECT r.ride_id, u.full_name AS rider, c.full_name AS captain, v.vehicle_type,
r.pickup_location, r.drop_location, r.fare, r.status FROM rides r JOIN users u ON r.user_id =
u.user_id JOIN captains c ON r.captain_id = c.captain_id JOIN vehicles v ON c.vehicle_id =
v.vehicle_id ORDER BY r.ride_id;
```

ride_id	rider	captain	vehicle_type	pickup_location	drop_location	fare	status
1	Arjun Mehta	Ravi Kumar	Bike	Koramangala	HSR Layout	85	Completed
2	Priya Sharma	Sunil Rao	Bike	Indiranagar	MG Road	120	Completed
3	Rohan Das	Deepak Joshi	Cab	Whitefield	Koramangala	340	Completed

ride_id	rider	captain	vehicle_type	pickup_location	drop_location	fare	status
4	Arjun Mehta	Imran Khan	Auto	HSR Layout	MG Road	95	Completed
5	Ananya Iyer	Lakshmi Bai	Cab	MG Road	Indiranagar	150	Completed
6	Vikram Singh	Manoj Gowda	Bike	Whitefield	HSR Layout	280	Completed
7	Sneha Reddy	Ravi Kumar	Bike	Koramangala	Indiranagar	70	Ongoing
8	Priya Sharma	Deepak Joshi	Cab	MG Road	Whitefield	360	Requested
9	Karan Patel	Sunil Rao	Bike	HSR Layout	Koramangala	90	Cancelled
10	Meera Nair	Lakshmi Bai	Cab	Indiranagar	Whitefield	310	Completed
11	Rohan Das	Manoj Gowda	Bike	HSR Layout	Whitefield	250	Completed
12	Ananya Iyer	Ravi Kumar	Bike	Koramangala	MG Road	100	Completed
13	Vikram Singh	Imran Khan	Auto	MG Road	Koramangala	80	Completed
14	Sneha Reddy	Sunil Rao	Bike	Indiranagar	HSR Layout	75	Completed
15	Meera Nair	Deepak Joshi	Cab	Whitefield	MG Road	330	Ongoing

Explanation: Inner joins follow foreign keys users ← rides → captains → vehicles. Every ride has a user, captain and vehicle, so 15 rows return. Cab trips (Deepak Joshi, Lakshmi Bai) show higher fares than typical bike trips, which matches vehicle capacity and distance.

3. Report B — Payment coverage (LEFT JOIN)

Finance must see which bookings still have no payment row.

```
SELECT r.ride_id, r.status, r.fare, p.payment_id, p.payment_method, p.payment_status, p.amount
FROM rides r LEFT JOIN payments p ON r.ride_id = p.ride_id ORDER BY r.ride_id;
```

ride_id	status	fare	payment_id	payment_method	payment_status	amount
1	Completed	85	1	UPI	Success	85
2	Completed	120	2	UPI	Success	120
3	Completed	340	3	Card	Success	340
4	Completed	95	4	Cash	Success	95
5	Completed	150	5	UPI	Success	150
6	Completed	280	6	Wallet	Success	280
7	Ongoing	70	NULL	NULL	NULL	NULL
8	Requested	360	NULL	NULL	NULL	NULL
9	Cancelled	90	NULL	NULL	NULL	NULL
10	Completed	310	7	Card	Success	310
11	Completed	250	8	UPI	Success	250

ride_id	status	fare	payment_id	payment_method	payment_status	amount
12	Completed	100	9	Cash	Success	100
13	Completed	80	10	UPI	Failed	80
14	Completed	75	11	UPI	Success	75
15	Ongoing	330	NULL	NULL	NULL	NULL

Explanation: LEFT JOIN keeps every ride. Rides 7, 8, 9 and 15 show NULL payment columns (ongoing, requested, cancelled, ongoing). Ride 13 is completed but payment_status is Failed — a collection follow-up case. An inner join would have hidden the unpaid rides, so LEFT JOIN is the correct report join.

4. Report C — Captains above average rating (subquery)

```
SELECT captain_id, full_name, rating FROM captains WHERE rating > (SELECT AVG(rating) FROM captains) ORDER BY rating DESC;
```

captain_id	full_name	rating
4	Deepak Joshi	4.8
1	Ravi Kumar	4.7
5	Lakshmi Bai	4.6

Explanation: The nested SELECT computes the fleet average (4.52). Deepak Joshi (4.80), Ravi Kumar (4.70) and Lakshmi Bai (4.60) qualify as premium captains for promotional badges. The subquery is independent of the outer row (non-correlated).

5. Report D — Riders who spent more than average (nested subquery + HAVING)

```
SELECT u.user_id, u.full_name, ROUND(SUM(p.amount),2) AS total_spent FROM users u JOIN rides r ON u.user_id = r.user_id JOIN payments p ON r.ride_id = p.ride_id WHERE p.payment_status = 'Success' GROUP BY u.user_id, u.full_name HAVING SUM(p.amount) > ( SELECT AVG(spend) FROM ( SELECT SUM(p2.amount) AS spend FROM payments p2 JOIN rides r2 ON p2.ride_id = r2.ride_id WHERE p2.payment_status = 'Success' GROUP BY r2.user_id ) ) ORDER BY total_spent DESC;
```

user_id	full_name	total_spent
3	Rohan Das	590
8	Meera Nair	310
5	Vikram Singh	280

Explanation: Successful payments are summed per rider. The inner query builds each rider's spend, then AVG(spend) is the benchmark. Rohan Das (₹590), Meera Nair (₹310) and Vikram Singh (₹280) sit above that benchmark — a loyalty / subscription target list. Failed payment of ₹80 is excluded, so Vikram's figure is the Whitefield bike trip only.

6. Report E — Captains operating in Koramangala (M:N join)

```
SELECT z.zone_name, c.full_name AS captain, v.vehicle_type, c.rating FROM zones z JOIN captain_zones cz ON z.zone_id = cz.zone_id JOIN captains c ON cz.captain_id = c.captain_id JOIN vehicles v ON c.vehicle_id = v.vehicle_id WHERE z.zone_name = 'Koramangala' ORDER BY c.rating DESC;
```

zone_name	captain	vehicle_type	rating
Koramangala	Deepak Joshi	Cab	4.8
Koramangala	Ravi Kumar	Bike	4.7
Koramangala	Imran Khan	Auto	4.2

Explanation: The junction table captain_zones resolves the M:N OPERATES IN relationship. Koramangala is covered by a cab, a bike and an auto captain, which is a healthy mix for that locality.

7. Report F — Revenue by vehicle type (join + GROUP BY)

```
SELECT v.vehicle_type, COUNT(r.ride_id) AS completed_rides, ROUND(SUM(r.fare),2) AS
total_revenue, ROUND(AVG(r.fare),2) AS avg_fare FROM vehicles v JOIN captains c ON
v.vehicle_id = c.vehicle_id JOIN rides r ON c.captain_id = r.captain_id WHERE r.status =
'Completed' GROUP BY v.vehicle_type ORDER BY total_revenue DESC;
```

vehicle_type	completed_rides	total_revenue	avg_fare
Bike	6	910	151.67
Cab	3	800	266.67
Auto	2	175	87.5

Explanation: Completed bike trips bring the largest total (₹910) because they are more frequent. Cabs earn more per trip (average ₹266.67 vs bike ₹151.67). Autos are fewest. This report supports fleet-planning decisions for Rapido Bengaluru.

8. Conclusion

Joins reconstruct the ER diagram at query time: identity stays in parent tables, facts stay in child tables. Subqueries compare a row or a group with a statistic computed from the same database. Together they produce the operational, finance and zone reports the Rapido project needs, using only the finalized relational design.